The Endocrinology of the Menstrual Cycle and its impact on the Nervous System's Emotion Centre.

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Abstract

Throughout this essay I will address the role of the brain and nervous system in regulating healthy female menstruation through its endocrinological mechanism. This will introduce the hormones oestrogen and progesterone; steroids which can cross the blood brain barrier and interact with neurotransmitters. Hence ovarian hormones collaborate with the limbic system affecting the ability to regulate mood. I discuss specific studies with examples where progesterone and oestrogen associate with key neurotransmitters, consequently affecting mood, relating back to instances during the menses as a potential cause. Finally I mention the case of oral birth control, its mechanism through artificial hormone ingestion as well as a brief overview on menstrual mood disorders.

Introduction

Neuroscientific research and progress in women's health has exponentially increased in the past two decades. This presents tremendous evolution to the scientific world, uncovering new hypotheticals and drawing links between previously unassociated topics. Accordingly, throughout this essay I aim to explore the recent ties between mood instability, menstrual phase and ovarian hormones fluctuations.

In today's society, there is a principal notion that a woman's biology can fluster her energy, attention, emotions and overall brain. A popular example claiming a 'moody' state of mind being tied with 'that time of the month'. This universal acknowledgment is only a comical glimpse of the depth ovarian hormones can reach throughout the nervous system. Menstruation's interaction with endocrinology is non-negotiable and acts as a control centre. However the steroids that fluctuate in regulation's returned impact on different neural circuits has been a recent area of increasing scientific intrigue. As women undergo the twenty-eight day cycle (during their fertile years), constant hormonal fluctuations driven by the nervous system create symptoms like acne, cravings, cramps and most notably mood instability. Recent studies show that there is a concrete scientific and neurochemical explanation to the experience of symptoms, particularly mood changes, during the different phases of menses as well as the general female lifespan.

Discussion

The brain and its mechanism

The human brain in its approximate three pounds is arguably the most remarkable processing machine. Being able to communicate through local and long range allows it to accommodate the entire human body. It is adapted to juggle an immeasurable volume of tasks to ensure species survival. These capabilities rest in the role of a web of neurons, interconnected nerve cells which allow instantaneous communication using electrochemical impulses throughout the whole organism.

In addition to providing intercellular communication throughout the body, neurons act as a gateway to interactions with the environment. Crucial in ensuring survival as it allows us to recognise external threats – removing our hand from a naked flame as we experience uncomfortable heat, and regulating internal and unconscious body mechanisms: the autonomic nervous system. The system receives information in the form of stimuli, both external and internal, processes it and sends out signals in the form of neurotransmitters to appropriately react[15]. It regulates involuntary physiological processes including digestion, respiration, heart rate, blood pressure and in women, the menstrual cycle usually through hormonal interactions[17–21].

Neurons are able to carry out their role because of the use of neurotransmitters: chemical messengers which help with the delivery of information. Neurotransmitters relay neuronal messages from target cells to receptors across gaps called synaptic clefts. Neurons relay messages as they ‘spike’ and fire an impulse releasing neurotransmitters from synapses.

Neuroimaging; Providing Data for Neuroscientific Advances

Throughout this discussion there are many mentions of different studies performed and results analysed through a variation of scans[8,9,10]. Hence, a brief overview of the different types of neuroimaging relating key hormones to brain functions and neurotransmitter mechanisms. Positron emission tomography (PET) and functional magnetic resonance imaging (fMRI) both help provide sufficient evidence in the form of neuroimaging as seen in figure 1. They have the potential to provide consistent and reproducible data for progress in terms of oestrogen or progesterone interactions with the brain.

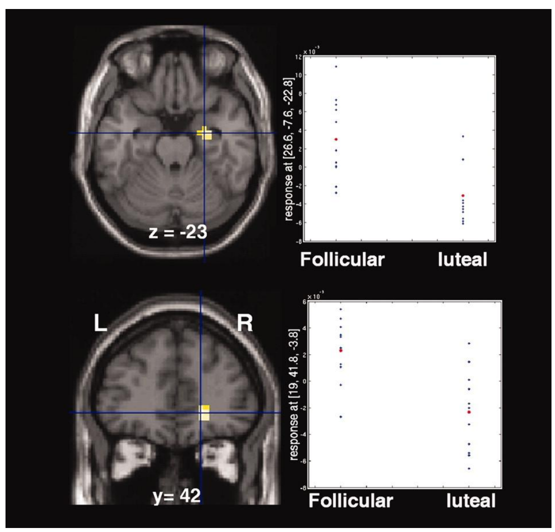

Figure 1: Statistical maps overlaid onto structural MRI showing BOLD fMRI responses greater in follicular phase than in luteal phase in the right amygdala (Upper) and orbitofrontal cortex (Lower)[60]

PET neuroimaging is based on the delivery, typically injected, of water or glucose paired with a radioactively labelled ligand: a tracer, to follow the direction of blood flow and main sites of glucose metabolism. The PET scanner allows for higher spatial resolution and improved sensitivity detection than alternative methods because it detects the emission of positrons from the administered substance from pairs of gamma rays as seen in figure 2[11].

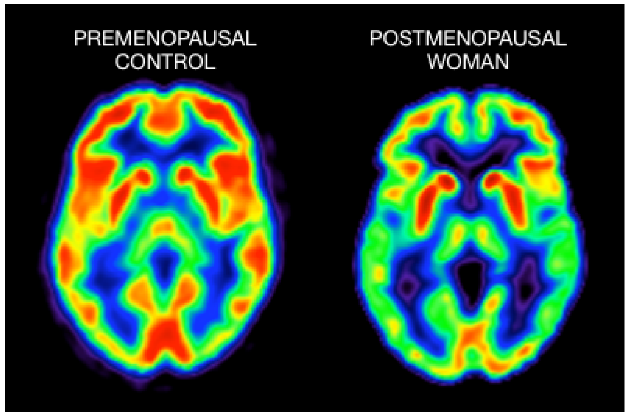

Figure 2: PET scans of a woman premenopausal (left) and postmenopausal (right) [59].

In contrast to PET scanning methods, fMRI's provide a noninvasive alternative to neuroimaging techniques. It works by displaying cerebral blood flow and measuring the difference in magnetic properties of haemoglobin, an oxygen carrier, regarding neuronal activity[13]. Hence, the lack of threat of exposure to excessive radiation, an fMRI session can undergo multiple tasks and extract more results from a single instance than PET scans. Yet despite its noninvasive appeal, one would opt for PET scans rather than fMRI when analysing the effects of oestrogen on areas of the brain like the hippocampus and frontal cortex. This is due to fMRI's sensitivity to air and bone artefact in medial temporal brain regions and air artefact in orbit and polar regions of frontal cortex[14]. Overall, both have been used to provide neural images that can therefore be studied and lead to scientific advances.

The Menstrual Cycle and its Hormones

The process required for healthy reproduction in women is stimulated and regulated by a complex combination of hypothalamic, pituitary and ovarian action using stimulatory and inhibitory signals. The four main hormones of the menstrual cycle fluctuate in concentration throughout each period as well as during reproductive milestones like menarche, pregnancy, and menopause. These are progesterone and oestrogen, released by the ovaries, alongside luteinizing hormone (LH) and follicle stimulating hormone (FSH) released by the pituitary gland. They prepare the uterus for pregnancy by contributing to the endometrium thickening, however if fertilisation does not occur then a woman undergoes the menstrual cycle, a four step process[1] which starts with the shedding of the endometrium.

These four phases are summarised into menstruation, the follicular phase, ovulation and the luteal phase[1]. The hypothalamus is a gland in the brain primarily made of grey matter which is involved in many aspects of the autonomic nervous system, and initiates the menstrual cycle. It acts as a metronome, setting the pace with the release of gonadotropin-releasing hormone (GnRH). The start of a woman's cycle is confirmed by the presence of blood, mucus and endometrial cells flowing out the vagina for an average of five days[2]. Menstruation happens in parallel with the follicular phase which in turn outlasts it, during a time frame of 13 to 14 days[2].

The follicular phase addresses the time frame when an egg matures in the ovaries in hope of fertilisation. Simultaneously, the pituitary gland formally signals the release of FSH stimulating the development of follicles as it gets triggered by GnRH. Follicles are fluid filled sacs in the ovaries, which once stimulated, secrete oestrogen, and consist of three main types of cells: theca cells, granulosa cells, and the oocyte. Theca cells interact with LH to produce androstenedione, a precursor for androgens, which is eventually converted into estradiol, a common form of oestrogen. This reaction is catalysed by aromatase, an enzyme synthesised by granulosa cells as they get stimulated by FSH. As the levels of oestrogen rise, a positive feedback mechanism occurs resulting in a rise of GnRH secretion and consequently a peak in LH levels. This initiates ovulation around day 14 of the cycle as seen in figure 3.

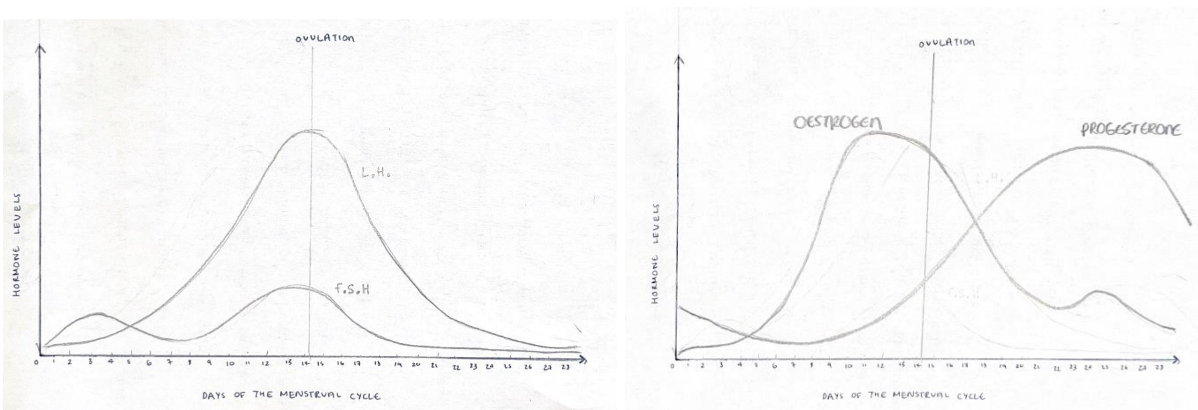

Figure 3: Graphs demonstrating the changes in levels of sex hormones throughout the menstrual cycle.

Ovulation occurs and a mature egg is released from an ovary, travelling along the fallopian tube to arrive at the uterus, over a 16 to 32 hours period[2]. The surge in LH is what enables the mature egg to be released and also presents the greatest chance for pregnancy[5,6]. Eventually oestrogen begins to inhibit FSH secretion, its levels decrease and oestrogen peaks once more as the follicular phase comes to an end. This cyclically increases the levels of LH – secreted by the pituitary gland, causing the formation of the vital temporary organ: the corpus luteum. The corpus luteum develops under the guidance of LH and the remnants of the dominant follicle as a consequence of the lack of fertilisation[7].

The release of GnRH is not restricted to the follicular phase and is also crucial to the 14 day luteal phase. This addresses the production of progesterone and a small amount of oestrogen from corpus luteum. The presence of these hormones prepare for pregnancy by thickening cervical mucus and thickening the uterine walls.[2] This prevents bacteria getting inside the uterus: an ideal environment for a fertilised egg to grow. If fertilisation does not occur, the end of the luteal phase is attained by a period as the endometrium sheds and the cycle begins once more.

The Limbic System and the Blood Brain Barrier

In the human brain, emotion and mood is controlled by the limbic system, a universally-debated list of structures. However the amygdala, hippocampus, limbic cortex and hypothalamus, all buried under the cerebral cortex, are non-negotiables when discussing emotion[22]. Feelings of anger, sadness, confusion and disgust, all get processed through these different interconnected areas of the brain. Despite its small size, the hypothalamus lies at the centre of the limbic system and is heavily involved in its mechanisms[23]. In particular, studies claim that fear responses are produced due to the paired hypothalamus and amygdala stimulation[24]. Alongside its role in the limbic system it also provides regulation towards hormone secretion and inhibition through the control of the pituitary gland.[25] Therefore because of this regulative role, it plays a key step in translating emotions into physical response usually by changing heart rate or signalling the release of hormones.

The pituitary gland interacts with the hypothalamus through the pituitary stalk, blood vessels and nerves, allowing impulses to travel across enabling communication. Together they form an aspect of the endocrine system: a network of glands that regulate subconscious body functions, one of which is sexual development[26]. As a key feature of female sexual development, the endocrine system is essential in stimulating hormones to trigger monthly bleeding. In respect to menstruation, the hypothalamus acts as a metronome as discussed above with the pulsed release of GnRH to then stimulate LH and FSH from the pituitary gland[27]. Oestrogen in return has a multifaceted effect on the emotional aspect of the hypothalamus, demonstrated through postpartum depression, premenstrual syndrome (PMS) or premenstrual dysphoric disorder

(PMDD). In addition, general depression and anxiety affect women in their oestrogen-producing years to a greater extent than postmenopausal women, or men[28,29,30].

Alongside their crucial role in menstrual regulation, both oestrogen and progesterone consist of an ability to pass through the blood brain barrier (BBB), due to their small size and lipid solubility. The BBB is a separation between the central nervous system and peripheral tissues allowing filtration of different chemical molecules. This protective layer of endothelial cells can be traversed either through direct trans-membrane diffusion or protein transporters. Since oestrogen and progesterone fall under the category of steroid hormones: secreted by a steroid gland “ovaries”, they cross the BBB via diffusion.

This removes the restriction on the reach of the hormone’s effects and explains the nervous system's sensitivity towards oestrogen and progesterone[16]. Hence, the hormones’ access to key neural pathways have been used to explain symptoms, changes in mental health and cognitive abilities during each cycle as well as throughout the average woman's life.

Oestrogen, Serotonin and Dopamine

As oestrogen levels fluctuate, studies have shown serotonin being impacted which trigger PMS symptoms[31]. Serotonin is a neurotransmitter, naturally found in the brain, that contributes to sleep, memory, hunger and happiness. It helps regulate a “feel good” state when levels are stable, with low levels associated with depression as seen in figure 5[32]. In fact many treatment plans for mood disorders are based on serotonin, such as a class of antidepressants known as selective serotonin reuptake inhibitors (SSRIs). Oestrogen increases the levels of serotonin and serotonin receptors in the brain as well as dopamine which correlates to an improvement in mood[33]. Hence we can infer that around day 12 and 22 of a woman’s cycle, an overall good-feeling is expected as oestrogen peaks in the mid-follicular and mid-luteal phase. Consequently, when there is a dip during the start of menstruation and after ovulation, a poorer mental state is expected.

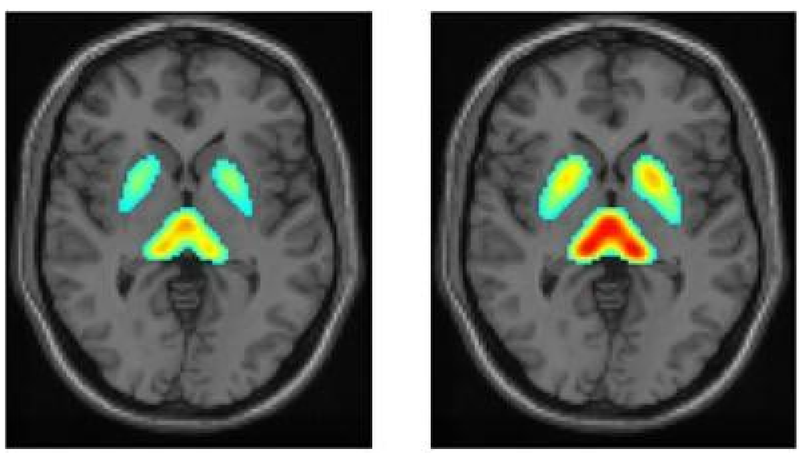

Figure 5: PET scan demonstrating reduced serotonin transporter (5-HTT) availability in the thalamus of patients with major depression (left) as compared to healthy controls (right)[32]

This is linked with oestrogen's contribution to the synthesis, reuptake and degradation of the serotonergic system. A study performed by Pecins-Thompson looked at the effect of ovarian steroids and raloxifene on proteins that synthesise, transport and degrade serotonin in the raphe region of oophorectomized and hysterectomised macaques[34]. His team treated the monkeys with oestrogen, oestrogen/progesterone and a control group in order to compare the presence of key enzymes which have a role in synthesis and metabolism of serotonin. Their data demonstrated a drastic increase in tryptophan hydroxylase mRNA: the rate-limiting enzyme in the synthesis of serotonin. The studies also consisted of a short term, 28 days, and prolonged treatment of five months. In comparison to the control group, the shorter treatment decreased the presence of serotonin reuptake transporter (SERT) whereas the extended treatment provided an increase in SERT (figure 6)[33]. More recent data also demonstrates the expression of SERT being both dependent on the presence of serotonin and the duration of hormonal exposure[36].

Applying the data provided from the Pecins-Thompson study, we can deduce that the 28 day treatment provided a decrease in SERT and therefore an increase in the firing of serotonin. This consequently contributes to a general positive mood and could further explain the negative emotions felt when there is a dip in oestrogen levels during certain phases of menstruation. This could particularly regard negative emotions towards the end of ovulation.

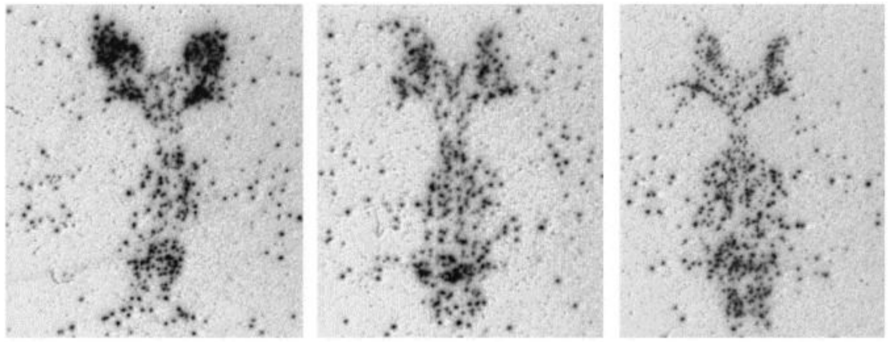

Figure 6: Autoradiographs from equivalent levels of the dorsal raphe obtained from three categories of monkeys showing a decrease in SERT mRNA signal as reflected by black pixels in the dorsal raphe of both E- and E+P-treated animals.[34]

Dopamine (DA), is a vivid chemical molecular component of the nervous system. It is both classified as a neurotransmitter and hormone - specifically a catecholamine[35]. As a catecholamine it is produced by the adrenal gland and contributes to bodily functions including memory, movement, motivation, arousal and the fight or flight response to name a few. Dopamine is known as a “feel-good” hormone as it provides a sense of pleasure and motivation. This is because it plays an active role in the body's reward system: an evolutionary adaptation to ensure survival. Hence low dopamine levels could lead to lack of motivation, mood swings alongside other generally negative mood-related sensations.

According to the Reproductive Health Research Institute[37], oestrogen was shown to increase the synthesis of dopamine alongside decreasing the degradation and reuptake. This promotes an increased “firing” of dopamine and hence, a greater sense of happiness and well-being[38]. We can apply this to stages of the menstrual cycle. Oestrogen levels are low before and during a period which explains key mood related symptoms. While mild discomfort and mood shifts prior and during menstruation have been recognised as classic symptoms, there are cases where a woman's day to day life is distinctly affected by the menses. These are PMS and PMDD which are classified as premenstrual disorders. Over the years, such disorders have gone by different names such as ‘menses moodiness' throughout the 1900’s and ‘premenstrual tension’ in the beginning of the 19th century[39]. Both PMS and PMDD are hormone-based disorders and therefore endocrine disorders of the luteal phase in menstruation. They are identified through behavioural symptoms that prevent female individuals from performing habitual tasks to the extent of impacting quality of life. Women that suffer from PMDD can develop depression, experience anxiety as well as suicidal thoughts all linked to a monthly drastic drop in oestrogen[40].

Progesterone, GABA and catecholamine

The endocrine system is an irreplaceable aspect of the human body, and equally important to a healthy female reproductive cycle[41]. It is a network of glands like the hypothalamus, ovaries and adrenal gland which produce and release hormones helping essential bodily mechanisms. Like all systems in the body, it is susceptible to disorders, characterised by either hormone imbalance or the development of a tumour. In the case of PMDD and PMS, they are a clear display of ovarian reproductive hormone imbalance[40]. Though the exact cause of PMDD is not yet known, the main theories revolve around oestrogen and progesterone. There have been recent advances linking it more directly to a gamma-aminobutyric acid (GABA) sensitivity specifically tied to unstable progesterone levels. In fact, allopregnanolone: synthesised by progesterone, under the commercial name ZULRESSO or Brexanolone[42], is prescribed to target GABA receptors to treat postpartum depression.

Progesterone, like oestrogen, is synthesised in ovaries and placenta in women, as well as glia and neurons in the nervous system making it a neurosteroid[43]. It is mostly present during the luteal phase of the cycle as well as pregnancy, stimulated by the pulsating release of LH. Similar to its counterpart oestrogen, progesterone also has a degree of power over mood swings and general emotional stability. This occurs because progesterone metabolises the neurosteroid allopregnanolone that acts as an allosteric modulator to GABA-A receptors[44]. It binds and activates the receptor cell stimulating the production of GABA, the primary inhibitory neurotransmitter in the brain.

GABA is crucial to inhibiting neuronal firing, used in many aspects of the nervous system and contributing everywhere in the body, one of which is in reproductive health as seen in the menstrual cycle. In addition GABA also corresponds to sleep and boosting your mood, feeling relaxed and overall good. Hence why pregnant women – who are seen to have extended peaks of progesterone for foetal health, report feeling “amazing” mentally during pregnancy[45,46]. This also explains why the sharp decline in progesterone levels after giving birth contribute to anxiety and postpartum depression treated through artificial allopregnanolone injections[42].

In addition to its link to GABA, progesterone impacts the female mood by regulating the secretion of catecholamine from chromaffin cells[47]. Chromaffin cells are a component of the adrenal medulla, the inner structure of the adrenal gland which control the stimulation of the ‘fight or flight’ response through hormone secretion. These hormones, including adrenaline noradrenaline, and epinephrine and dopamine, fall under a category of hormones called catecholamines (CA). Stress stimulates the secretion of CAs alongside progesterone, which in turn is hypothesised to regulate stress-induced CA release in a 2005 study on bovines. Siobhan Armstrong and Edward Stuenkel used bovine chromaffin cells to investigate the potential effect progesterone could have on CA secretion. They provided evidence confirming that in chromaffin cells, progesterone strongly and powerfully inhibited the nicotinic acetylcholine receptor (nAChR) which consequently affected the immediate inhibition of CA secretion[49]. The nAChR being a receptor that responds directly to acetylcholine: an excitatory neurotransmitter and a vital component of memory formation and recall[20].

Artificial Hormone use: Birth Control and Treatments

A method of contraception by an estimated 151 million women due to efficiency and ease of use is oral contraception[51,52]. While physical implications of prolonged use of birth control have been validated with data, the psychological consequences are only recently being addressed[57]. Hence, growing intrigue towards oestrogen and progesterone's influence on the brain's capacity to regulate mood through oral contraceptives[50]. Majority of women of reproductive age use the pill as a way to avoid unwanted pregnancy, however 14% for other reasons like acne or irregular periods [51,52,53].

The' oral pill contains a low dose of hormones, with its mechanism of action revolving primarily around progesterone's ability to prevent ovulation. Progesterone inhibits follicular development and prevents ovulation through a negative feedback interaction with the hypothalamus, decreasing the pulsed release of GnRH. As a response to the decrease in GnRH, the secretion of FSH and LH simultaneously decreases. This occurs as there is no follicle secreting estradiol, preventing an expected spike. Eventually, the lack of oestrogen positive feedback alongside the progesterone negative feedback on LH secretion halts the menstrual course mid-LH spike. Pregnancy is therefore prevented as ovulation is averted.[51,55].

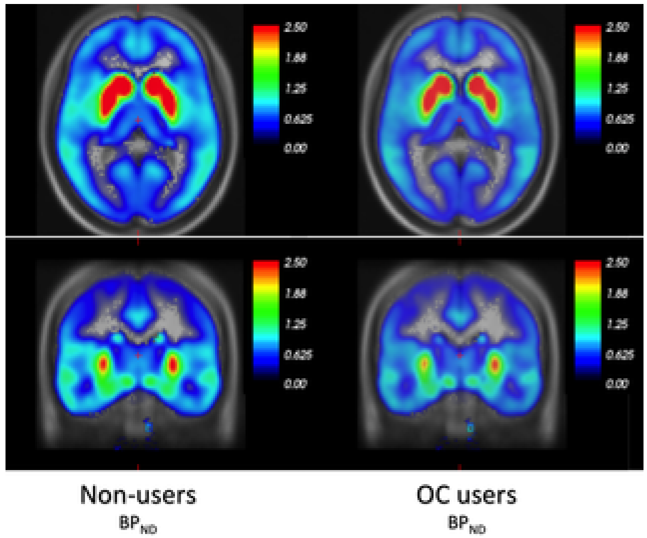

Figure 7: PET scan Illustration of the global effect of OC use on 5-HT4 serotonin receptors. OC users appear paler both in subcortical and cortical areas.[61]

A Danish study also explored the use of oral hormonal contraceptives and its association with depression across a range of age groups. The study on over one million women provided data demonstrating an increased risk of depression across all age groups, but notably among teenagers, when using any type of hormonal contraceptive[56]. Recently there has been a rise in population-based cohort studies with one of the largest led by researchers at Uppsala University in Sweden. Their results demonstrated a 130% higher incidence of depression symptoms in women who, as adolescents, used contraceptive pills. As well as a 71% higher risk of developing depression during the first two years using these hormonal pills[56].

In contradiction, a study published in 2021 by the British Journal of Obstetrics and Gynaecology, based on 39,585 women aged 15–25 years disagreed with these claims. It demonstrated that the participants who were taking combined oral contraceptives as well as oral progestogen-only products had lower or no increased risk of depression”[58]. They concluded this by comparing non-users, women on combined oral contraceptives and oral progestogen-only products. The results: “did not find any associations between use of COCs [combined oral contraceptives], which is the dominating HC [hormonal contraceptive] in first time users, and depression. Non-oral products were associated with increased risks.”[58]. Hence there was no grounds for association with oral contraceptives and depression or similar symptoms.

Conclusion

Over the past decade, neuroscientific research has progressively ventured into the female-oriented world of menstruation. The four phases of the menstrual cycle each have particular hormone levels to allow a healthy reproductive arc. These fluctuations have been seen to impact mood and mental wellbeing and by extension the limbic system. As the limbic system works to regulate emotions, key neurotransmitters are involved like serotonin, GABA and various catecholamines. This data has been used to create hormone based treatment like Brexanolone and SSRIs to address particular mood disorders which have been known to stem from neurotransmitter variations. This is important because it helps tie mood instability and menstrual phases with their relationship to neurotransmitter interactions. However there are also instances where artificial hormone use through birth control has shown adverse effects on mood, promoting depression - but oral contraceptive has provided lots of conflicting evidence and the studies are not as reproducible. Hence using the studies above I conclude that there is a clear relationship between the menstrual cycle's endocrinology and the nervous systems emotion centre.

Evaluation

In the world of science, experiences of bias, reproducible struggles and general inconsistencies in findings prevent simple and rapid progress towards any particular topic. Regarding the studies I have mentioned throughout my discussion, many have not been performed on humans but rather rats, monkeys or bovine. This would therefore limit the applicability of each study in regards to humans and specifically the female brain and nervous system. In regards to inconsistencies in findings towards the menstrual cycle, this could include issues surrounding the time frame of such assessments, lack or incorrect confirmation of cycle phases or lack of methodological standardisation. To improve the relevance of my essay I ensured the majority of the studies used were from the past decade this greatly ensures they are up to date with upcoming research and applicable.

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